length and duration—become irrelevant or meaningless. Just as you can sensibly study the temperature and viscosity of liquid water—concepts that apply to the macroscopic properties of a fluid—but when you get down to the scale of individual H_2O molecules, these concepts cease to be meaningful, so, perhaps, although you can divide regions of space and durations of time in half and in half again on everyday scales, as you pass the Planck scale they undergo a transformation that renders such division meaningless.

Many string theorists, including me, strongly suspect that something along these lines actually happens, but to go further we need to figure out the more fundamental concepts into which space and time transform.[33] To date, this is an unanswered question, but cutting-edge research (described in the final chapter) has suggested some possibilities with far-reaching implications.

The Finer Points

With the description I've given so far, it might seem baffling that any physicist would resist the allure of string theory. Here, finally, is a theory that promises to realize Einstein's dream and more; a theory that could quell the hostility between quantum mechanics and general relativity; a theory with the capacity to unify all matter and all forces by describing everything in terms of vibrating strings; a theory that suggests an ultramicroscopic realm in which familiar space and time might be as quaint as a rotary telephone; a theory, in short, that promises to take our understanding of the universe to a whole new level. But bear in mind that no one has ever seen a string and, except for some maverick ideas discussed in the next chapter, it is likely that even if string theory is right, no one ever will. Strings are so small that a direct observation would be tantamount to reading the text on this page from a distance of 100 light-years: it would require resolving power nearly a billion billion times finer than our current technology allows. Some scientists argue vociferously that a theory so removed from direct empirical testing lies in the realm of philosophy or theology, but not physics.